The Shape of Wisdom: How Language Models Converge on an Answer

Shailesh Rana
Independent Researcher

Abstract

When a language model answers a factual question, where does its internal state go? We asked four open models (Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B and Llama-3.2-1B) 2,256 factual questions whose answers recur, such as the country a person is a citizen of, and recorded the state from which each answer is produced at every layer. Questions that receive the same answer converge on a shared region of state space, which we summarise by its centre, the *answer centroid*. Our findings are four. First, the answer becomes readable in a single layer: in every model, the share of questions whose state lies nearest its own answer's centroid jumps within one layer, at a depth between 47% and 71% of the way through the network. Second, centroids follow the model's answer, not the question or the truth: the centroid of a country learned from questions about people's citizenship also identifies that country in questions about towns and films. Third, the model uses the centroids: after the jump, moving the state of a wrongly answered question to the right answer's centroid makes the model give the right answer in 83%–97% of cases late in the network, against 2%–4% for a random move of the same size, and before the jump such moves do nothing. Fourth, centroids reveal near misses: when a model answers wrongly but the right answer's centroid is among the nearest to its state, the model usually picks the right answer when the question is asked as multiple choice. This signal separates near misses from real gaps in knowledge better than the model's confidence does, in all four models.

1 Introduction

Ask a language model "What country is Aníbal Ponce a citizen of?" and, a few dozen layers later, it answers "Argentina". On the way, the model's internal state at the last position of the prompt, from which the answer is produced, changes at every layer. This paper asks what shape that change has. Does the state wander, drift slowly toward the answer, or arrive somewhere definite? And if it arrives somewhere, is it the same somewhere for every question with the same answer?

It is. As a question passes through the network, its state moves into a region that it shares with every other question to which the model gives the same answer. Figure 1 shows this in Llama-3.1-8B. Early in the network, questions whose answers are twelve different countries form one tangled cloud. Within a few layers, most of them separate into one strand per country.

To study this, we built 2,256 factual questions from Wikidata whose answers recur: 62 countries, cities, languages and sports, each the answer to many questions. We asked them of four open models and recorded the state at every layer. For each answer and layer, we computed the *answer centroid*: the average state of the questions to which the model gave that answer. We then measured, layer by layer, how often a question's state lies nearest its own answer's centroid, tested what the centroids encode, and moved states between centroids to see whether the model uses them. We find four things.

The answer becomes readable in a single layer (Section 3.1). In each model, the share of questions whose state lies nearest its own answer's centroid jumps within one layer, at a depth that differs between models.

Centroids follow the model's answer, not the question or the truth (Section 3.2). The centroid of a country learned from questions about people also identifies that country in questions about towns and films. It survives rewording, and it does not depend on the directions the model uses to write the answer word. When two models disagree, each state follows its own model's answer.

The model uses the centroids (Section 3.3). Moving a state to an answer's centroid after the jump makes the model give that answer. Random moves of the same size rarely do, and moves made before the jump have no effect.

Centroids reveal near misses (Section 3.4). Some wrong answers are near misses: shown the options, the model picks the right one. In a near miss, the right answer's centroid is among the nearest to the state. This tells near misses apart from real gaps in knowledge better than the model's confidence does.

2 Setup

This section covers what every result relies on: the questions, the models and their recorded states, the answer centroids, and readability. Each result in Section 3 then describes its own test and controls. Details needed only for reproduction are in Appendix A.

2.1 Questions

To see whether questions with the same answer reach the same state, we need many different questions for each answer. We

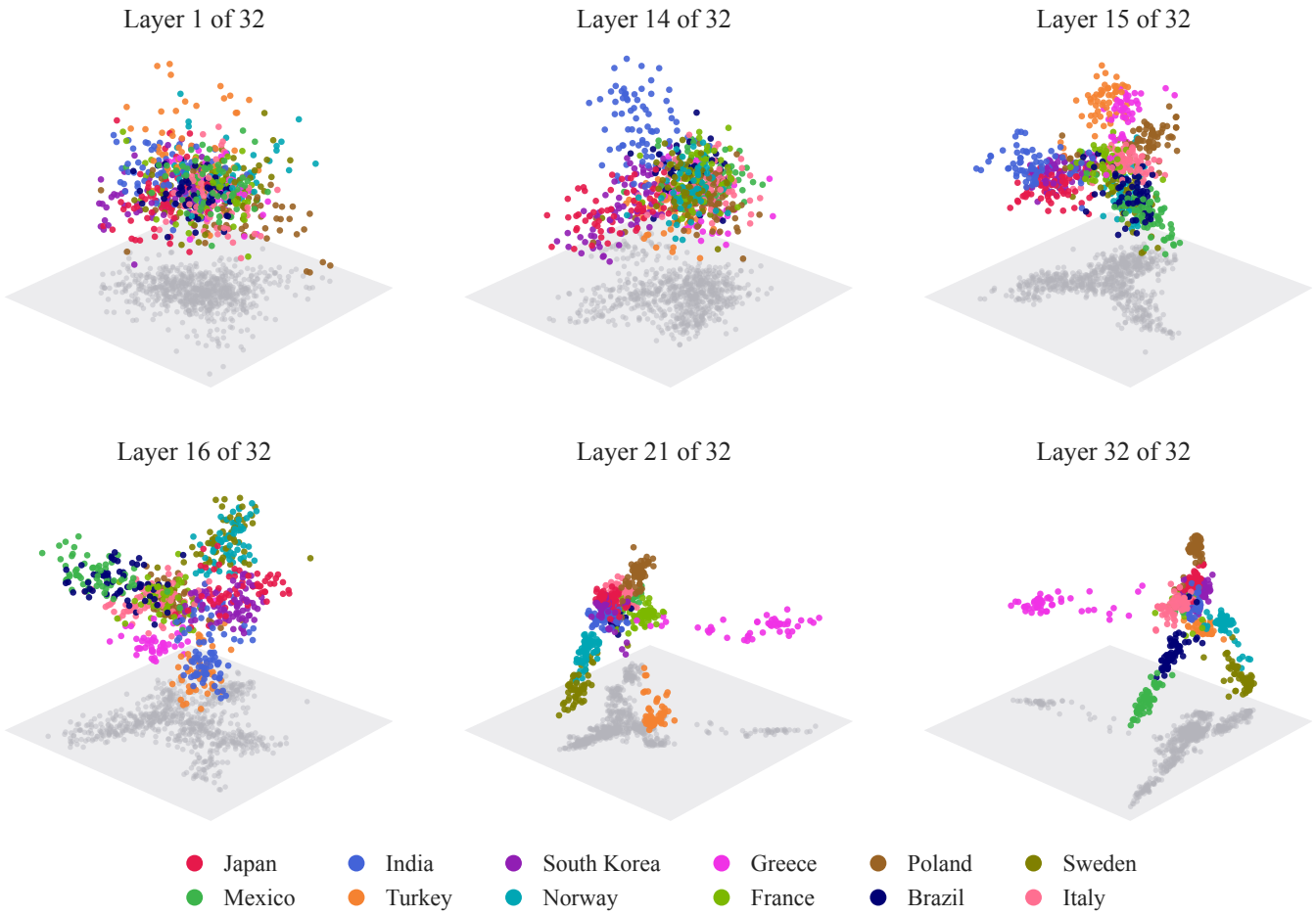


Figure 1: States of questions at six layers of Llama-3.1-8B. Each point is one test question about a person, town or film whose answer is one of the twelve countries the model gave most often; colour is the country the model answered. Each panel shows the questions' states at one layer, projected onto the three directions that best separate the twelve countries among the training questions at that layer (found by linear discriminant analysis, LDA). At layer 1 all questions share one cloud, and at layer 14 a few countries, such as India, stand apart. At layer 15, where readability jumps (Section 3.1), most countries separate; by the last layer most form a strand of their own. Each panel is scaled separately; grey dots are shadows on the floor.

Answer	Relation	Plain question
Country	Citizenship	What country is X a citizen of?
Country	Town's country	In which country is X?
Country	Film's country	Which country is the film X from?
City	Birthplace	In which city was X born?
City	Headquarters	In which city is X headquartered?
Language	Language spoken	Which language did X speak or write?
Language	Film's language	What is the original language of the film X?
Sport	Sport played	Which sport does X play?

Table 1: The eight relations. X is the subject: a person, town, film, company or athlete.

built such a set from Wikidata [36]: 2,256 facts with 62 possible answers of four kinds, namely 24 countries, 16 cities, 12 languages and 10 sports. The facts come in eight types, which Wikidata calls *relations* (Table 1). Several relations share a kind of answer. A country, for example, can answer a question about a person's citizenship, about where a town is, or about where a film is from. The subjects range from famous to obscure, so the models know some facts and not others.

Each fact is asked as an open question in up to three wordings: the plain question of Table 1, and two rewordings in different styles, written by a language model that was never told the answer. For example, a casual rewording of the plain question "What country is Aníbal Ponce a citizen of?" is "Hi, I'd appreciate knowing which country Aníbal Ponce is a citizen of, if possible?" The plain question is also asked as multiple choice with four options; this format is used only in Section 3.4.

2.2 Models and states

We study four instruction-tuned models: Llama-3.1-8B-Instruct [12], Qwen2.5-7B-Instruct [32], Gemma-2-9B-it [7] and Llama-3.2-1B-Instruct. Each answers greedily. At the last position of the prompt, where the answer is produced, we record the residual stream after every layer. We call this vector the question's *state* at that layer. *Depth* is the fraction of the model's layers that the state has passed through, from 0 at the input to 1 at the output.

A reply is right or wrong if it names exactly one answer;

replies that hedge, refuse or name two answers are set aside (0.2%–8.7% of replies, depending on the model).

2.3 Answer centroids

An answer's *centroid* at a layer is the average state, at that layer, of the questions to which the model gave that answer:

$$c_a = \frac{1}{|Q_a|} \sum_{q \in Q_a} h_q, \tag{1}$$

where

- a is an answer, for example Argentina;
- Q_a is the set of training questions (defined below) that the model answered with a, and $|Q_a|$ is their number;
- h_q is the state of question q at the layer;
- c_a is the centroid of a at that layer.

The centroid of Argentina at layer 15 of Llama-3.1-8B, for example, is the average layer-15 state of all training questions that Llama answered with "Argentina", whether they asked about a person, a town or a film, and in any wording. Centroids are built from the answers the model gives, not from the true answers. A question that the model wrongly answers with "Chile" counts toward the centroid of Chile, so wrong answers can be studied in the same way as right ones.

Centroids are built from questions about one half of the subjects, the *training questions*. Every number we report is measured on questions about the other half, the *test questions*. We split by subject, not by question, because the several questions about one subject are near copies of each other.

2.4 Readability

Readability measures how often the answer a model gave can be identified from its state alone. For each test question, we find the centroid nearest to its state, among answers of the same kind:

$$\hat{a}_q = \arg\min_{a \in A_q} \big\| W(h_q) - W(c_a) \big\|, \tag{2}$$

where

- h_q is the state of test question q at the layer;
- A_q is the set of answers of the same kind as the answer the model gave to q, for example the 24 countries for a country question;
- c_a is the centroid of answer a at the same layer;
- W is a whitening, described below;
- $\|\cdot\|$ is the ordinary straight-line distance;
- $\hat{a}_q$ is the answer whose centroid is nearest.

Readability at a layer is the share of test questions for which $\hat{a}_q$ is the answer the model actually gave. If states carried no information about answers, readability would be about one over the number of candidate answers: 4% for countries, 10% for sports.

Distances are measured after whitening because a few dimensions of the residual stream take much larger values than the rest and would otherwise decide every distance. The whitening W keeps the 128 principal components of the training states at that layer (found by principal component analysis, PCA) and rescales each to unit variance, so that every direction counts equally.

3 Results

3.1 The answer becomes readable in a single layer

If the model formed its answer gradually, readability would rise smoothly with depth. It does not. In every model the curve has the same shape: a low plateau, a jump within a single layer, and a high plateau (Figure 2A). In Llama-3.1-8B, readability rises from 32% at layer 14 to 71% at layer 15. The other three models show the same one-layer jump (Table 2). After the jump, readability keeps rising to a peak of 86%–94%. We call the layer of the jump the *onset layer*: the first layer at which readability covers half the distance from chance to its peak. To choose it without test questions, we measure this readability on training questions, building centroids from one half of the training subjects and testing on the other (Appendix A). The depth of the onset layer, the *onset depth*, differs between models, from 47% to 71% of the way through the network.

Two controls show how much readability our procedure reaches when the state carries no information about the answer. *Shuffled answers* keeps every state but credits each question with the answer of a different, randomly chosen fact of the same kind; readability then measures pure chance, which is low because there are 10 to 24 candidate answers of each kind (6%–8% across models). *Question text only* replaces the model's state with a vector built from the question's spelling alone, without the model.[1] It asks how far the wording gives the answer away, for example a name that sounds Japanese. It reaches 11%–18%. Readability above both lines is information that the model added to the state. Before the jump, readability is already 20%–55%, above both controls; this early signal may come from cues in the subject's name, which we did not test, and it does not grow toward the jump. The jump is much smaller for sports, the one kind of answer that is not a name: 7–19 percentage points, against 41–52 for countries, cities and languages (Section 5).

Readability depends on the whitening. To check the jump in the model's own states, we also measure *separation*: how far apart the centroids are, compared with how widely questions scatter around their own centroid. At each layer,

$$S = \frac{D_{\text{between}}}{D_{\text{within}}}, \tag{3}$$

[1] The vector counts every sequence of two to four characters in the question, weighted so that sequences common to many questions count less (term frequency–inverse document frequency, TF-IDF), and is compressed to 256 numbers.

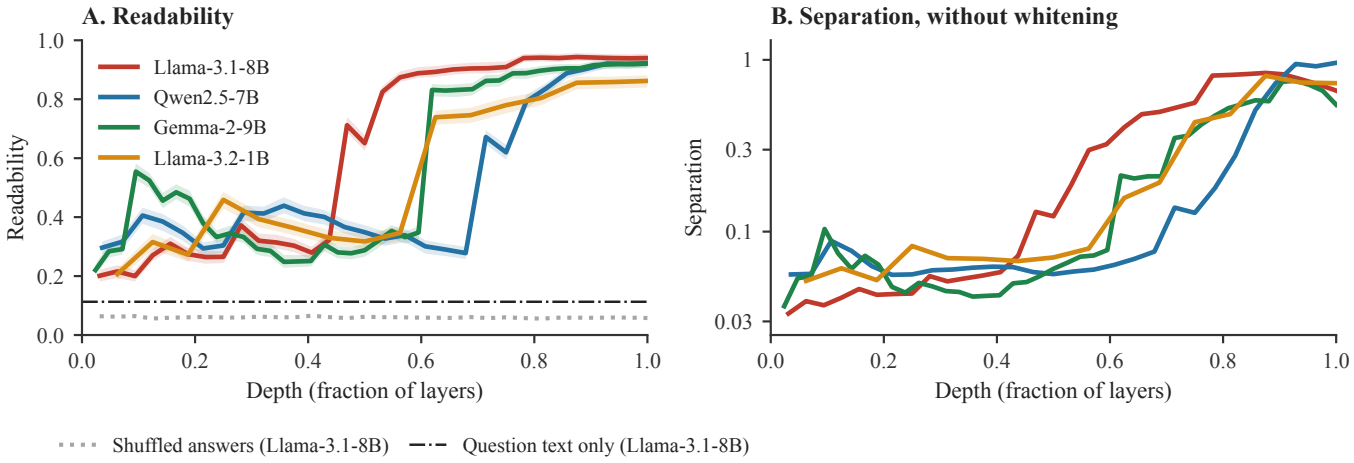


Figure 2: Readability and separation by depth. (A) Readability for the four models. The two horizontal lines are the controls, shown for Llama-3.1-8B; the other models' controls are similar. (B) Separation for the four models, on a log scale. Bands are 95% intervals.

with

$$D_{\text{between}} = \frac{1}{|A|} \sum_{a \in A} \|c_a - \bar{c}\|^2,$$

$$D_{\text{within}} = \frac{1}{N} \sum_{q} \|h_q - c_{g_q}\|^2,$$

where

- A is the set of answers of one kind, $|A|$ is their number, and $\bar{c}$ is the average of their centroids;
- D_{between} is therefore the average squared distance of a centroid from the centre of all centroids: how far apart the centroids are;
- the sum in D_{within} runs over the N test questions of that kind, and g_q is the answer the model gave to question q;
- D_{within} is therefore the average squared distance of a question from its own answer's centroid: how widely questions scatter;
- all distances are measured on the unwhitened states.

Higher separation means questions sit closer to their own centroid relative to the distance between centroids. We combine the four kinds of answer by weighting each by its number of test questions.

Separation shows the same event (Figure 2B). From its lowest to its highest value it rises 16- to 25-fold, and the rise is concentrated at the onset. The two layer steps at the onset, into the onset layer and out of it, carry 16%–32% of the whole rise on a log scale. If the rise were spread evenly over a model's 16 to 42 layers, two steps would carry only 5%–13%. Separation then keeps rising more slowly for about ten layers. This differs from trained classifiers, in which classes separate by a roughly constant factor at every layer [17].

3.2 Centroids follow the model's answer, not the question or the truth

A centroid is built from questions that share an answer. It could still reflect something else those questions share: the type of fact asked, the wording, the word the model is about to write, or the true answer. We test each possibility by changing how the centroids are built or what the state contains, and measuring readability again (Figure 3, Table 2). Below, "standard" readability is that of Section 2.4; for Llama-3.1-8B it is 94% at 90% of depth.

	Llama 8B	Qwen 7B	Gemma 9B	Llama 1B
Layers	32	28	42	16
Accuracy	63%	58%	67%	48%
Onset layer	15	20	26	10
Onset depth	47%	71%	62%	62%
Readability				
Layer before onset	32%	28%	35%	35%
Onset layer	71%	67%	83%	74%
Peak	94%	92%	92%	86%
Standard, 90% depth	94%	90%	92%	86%
Other relations	91%	85%	90%	76%
Other wordings	94%	90%	92%	85%
Answer words removed	94%	90%	91%	85%
True-answer centroids	75%	65%	73%	61%
Telling near misses from gaps (AUC)				
Rank of right centroid	0.77	0.75	0.75	0.82
Confidence	0.66	0.53	0.49	0.66

Table 2: Per-model summary. Accuracy: share of plain test questions answered correctly. Readability rows from "Standard" down are at 90% of depth; the four below "Standard" are the tests of Section 3.2. AUC: Section 3.4.

Other relations. If centroids were tied to the type of fact, a country centroid built from questions about people would not identify that country in questions about towns or films. We built centroids from the training questions of one relation only, measured readability on test questions of the other relations with the same kind of answer, and averaged over the choice of relation. (Sports have a single relation and are left out.) In Llama-3.1-8B, readability at 90% of depth is 91%, against 94% with standard centroids on the same kinds of answer. The centroid of a country is nearly the same whether the question is about a person, a town or a film.

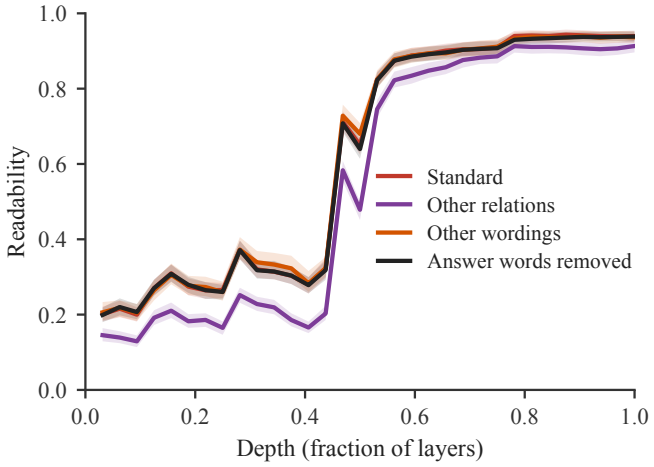


Figure 3: Readability of Llama-3.1-8B under three changes (Section 3.2). The standard, other-wordings and answer-words-removed curves nearly coincide. Bands are 95% intervals.

Other wordings. If centroids were tied to the wording, centroids built from plain questions would fail on rewordings such as the casual example in Section 2.1. We built centroids from plain questions only and measured readability on the rewordings. It is 94%, the same as standard.

Answer words removed. A centroid could simply reflect the model getting ready to write a particular word, such as "Argentina", which every question answered with Argentina shares. The model writes a word through the unembedding vectors of its tokens. We removed from every state its component along the unembedding vectors of all tokens of all 62 answer names, then rebuilt the centroids and measured readability again. It stays at 94% in Llama-3.1-8B, unchanged, and removing the same number of random directions does not change it either. The answer can therefore be read from the state without the directions that write the answer word.

True answers. If centroids reflected the true answer, centroids built from true answers would work at least as well as centroids built from the answers the model gives. They work worse: at 90% of depth they identify the true answer in 61%–75% of test questions, against 86%–94% for standard centroids identifying the given answer. Two models that disagree show the same thing. Take the plain test questions that two models answer differently. Each model's state lies nearer its own centroid for its own answer than its own centroid for the other model's answer in 66%–88% of these questions, across all 12 ordered pairs of models.

3.3 The model uses the centroids

Readability shows that the answer can be read from the state. It does not show that the model uses this information. If it does, moving a state to another answer's centroid should change the answer. For each test question that the model answered wrongly (121–233 per model, those whose wrong answer has a centroid), we changed its state at one layer to

$$h' = h + (c_{\text{right}} - c_{\text{given}}), \tag{4}$$

where

- h is the question's state at that layer;
- c_{given} is the centroid of the wrong answer the model gave;
- c_{right} is the centroid of the right answer;
- h' is the moved state, which replaces h while the model reads the prompt; the model then writes its reply as usual.

The centroids are the training centroids of Section 2.3 at that layer, and the move is made in the model's own, unwhitened states. We made this move at one layer at a time, at seven depths from 20% to 90% of the network. As controls, we moved the state by the same distance in a random direction, and toward the centroid of a third, randomly chosen answer of the same kind. We also moved right answers toward another answer's centroid.

Moves work only after the onset (Figure 4). At 20% of depth they correct 0%–1% of wrong answers. After the onset, the later the move, the more often it works. At 90% of depth, it corrects 97% of wrong answers in Llama-3.1-8B, 94% in Qwen2.5-7B, 97% in Gemma-2-9B and 83% in Llama-3.2-1B, while random moves of the same length give the right answer in only 2%–4% of cases. The effect is not specific to the truth. At 90% of depth, moves toward a third answer's centroid produce that answer in 82%–95% of cases, and right answers moved toward another answer's centroid switch to it in 68%–89%.

Because centroids move apart with depth, the moves grow from 3%–5% of the state's length at 20% of depth to 42%–49% at 90%, so late moves may partly write the answer directly. The clearest evidence is therefore the depth at which moves start to work, and the failure of random moves of the same length.

3.4 Centroids reveal near misses

Some wrong answers are *near misses*: the model gets the open question wrong but picks the right option when the same question is asked as multiple choice. We count a wrong answer as a near miss only if the model picks the right option in at least three of the four option orders. This holds for 60% of wrong answers in Llama-3.1-8B, 54% in Qwen2.5-7B, 52% in Gemma-2-9B and only 6% in Llama-3.2-1B. We call the other wrong answers *gaps*. Can the state tell the two apart, and can the model's confidence?

For each wrong answer to a plain test question, we computed two numbers. The first is the *rank of the right centroid*. We use the onset layer, fixed in advance as the first layer at which answers are readable; much later, nearly every state lies nearest the centroid of the answer the model gave. At that layer, we rank the centroids of all answers of the question's kind by their distance from the state, with 1 for the nearest, and take the rank of the right answer's centroid. The second is the model's *confidence*: its average log-probability per token

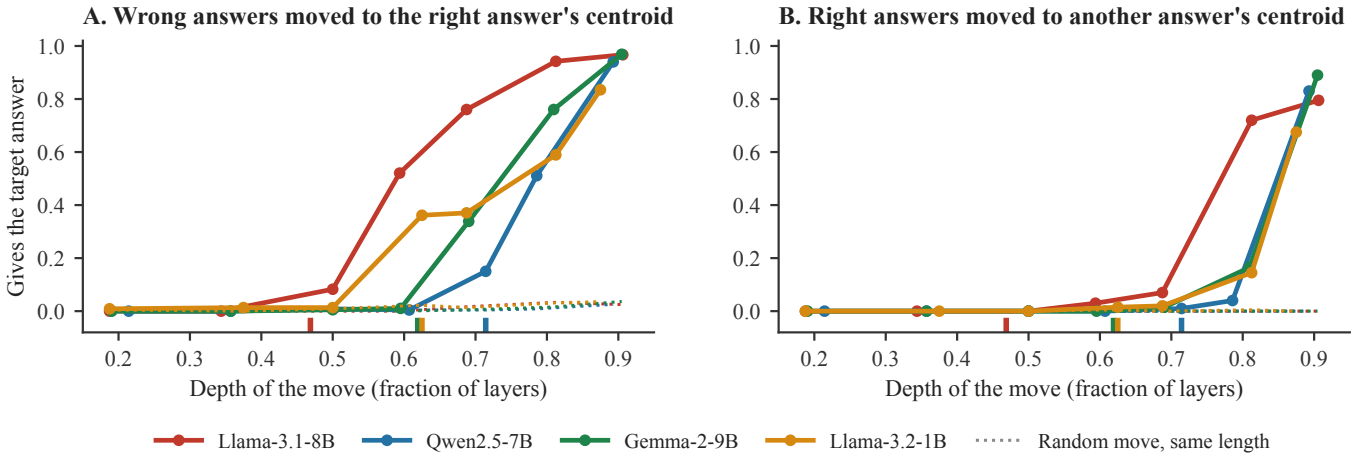


Figure 4: Moving states to answer centroids. (A) Wrong answers moved to the right answer's centroid. (B) Right answers moved to another answer's centroid. Solid lines: moves to the centroid; dotted lines: random moves of the same length. The y-axis is the share of questions for which the model then gives the target answer. Short bars under each axis mark each model's onset depth.

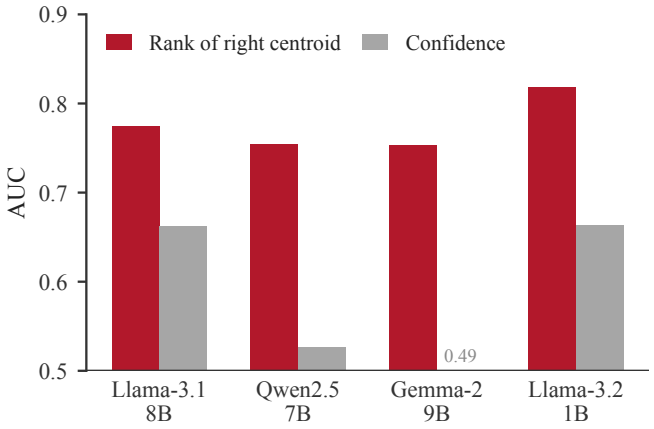


Figure 5: Telling near misses from gaps. AUC of each number among wrong answers. The axis starts at chance (0.5), so Gemma-2-9B's confidence (0.49) has no bar.

of its own reply. For each number, we measure how well it separates near misses from gaps by the area under the receiver operating characteristic curve (AUC): the probability that a randomly chosen near miss scores better than a randomly chosen gap. An AUC of 0.5 is chance and 1 is perfect separation. Neither number is fitted to the data.

The rank of the right centroid separates near misses from gaps better than confidence in every model (Figure 5): AUC 0.75–0.82, against 0.49–0.66. The 95% interval for the difference excludes zero in every model. For near misses, the right answer's centroid has a median rank of 1–2, depending on the model; for gaps, 4–8. A near miss is thus visible inside the model: the right answer's centroid is among the nearest to the state, often just behind the centroid of the answer the model gave. The model's confidence barely shows this.

4 Related work

Reading intermediate layers. The logit lens [26] decodes an intermediate state through the model's output layer, and the tuned lens [1] learns a decoder for each layer. Both show predictions sharpening with depth. Future Lens [28] and Patchscopes [10] read richer content from a single hidden state. All of these read the state through the output vocabulary. Our centroids are fitted from the states themselves, and reading them does not need the answer words' output directions.

How depth is used. Early layers gather and detokenise, middle layers build abstract features, and late layers refine a prediction [3, 4, 13, 20, 35]. Correct and incorrect predictions follow similar confidence curves across layers [19]. In factual recall, information about the relation and then the subject moves to the last position, where attention heads extract the attribute [9]. Feed-forward layers act as key–value memories [8], and middle-layer feed-forward modules at the subject's last token are causally decisive [24]. Our onset depth marks, for a whole population of questions, the depth at which each model's answer becomes available at the last position.

Representation geometry. Middle layers represent the same content in a shared space across languages and modalities [37, 41]. Concepts have structured geometry. Categories form simplices [30], some features are circular [5], space and time are linearly decodable [14], counts lie on curved manifolds that attention reshapes [15], and truth has a linear direction [23]. In trained classifiers, representations of different classes separate by a roughly constant factor per layer [17]. Language models move partway toward neural collapse, in which states gather at their class means, at their top layer, and further with scale [40]. The residual stream contains stable regions, within which small changes leave the next-token prediction unchanged [18]. Yet, almost surely, no two prompts map to the same last-position state [25]. Centroids fit both findings: questions can gather in the directions that carry the answer while staying distinct in others.

What models know but do not say. Internal states often encode more than the output shows [6, 27, 33]. On knowl-

edge benchmarks such as MMLU and ARC, however, this gap is near zero [31], and internal states mostly reflect what the model recalls rather than what is true [2]. False few-shot demonstrations can override a correct internal answer at a critical layer [16], and users' suggestions shift answers [34]. In multiple choice, attention in a few middle layers selects the answer letter, and a sparse set of heads then raises its probability [38]. The winning option is chosen by its content before it is bound to its letter [39], and models carry a prior over letters [42].

Steering. Adding a direction to the residual stream changes a model's behaviour [21, 29]. Our moves are of this kind. Each one, however, takes a single question's state from one measured centroid to another at one layer, and is compared with random moves of the same length.

5 Limitations

The questions are short and factual, and their answers come from a closed set of 62 countries, cities, languages and sports. Three of these four kinds are names. For the one kind that is not, sports, the jump is much smaller (Section 3.1) and readability at 90% of depth is lower (63%–90% against 89%–94%). Sports have at most ten answers, all from one relation, so we cannot tell whether this reflects the kind of word or the smaller set. Places for open-ended answers, multi-step reasoning or long contexts may look different.

We studied four instruction-tuned models between 1 and 9 billion parameters, with one prompt format each and greedy answers. Readability and separation both measure where states sit, not the computation that moves them there. Readability jumps within one layer, but separation keeps rising for about ten layers after the onset, so the convergence itself is not instantaneous. Moves at late layers may partly write the answer directly, so the strongest causal evidence is the depth at which moves start to work and the failure of random moves.

Wikidata answers are a reference, not ground truth. We removed one unreliable relation–answer pair after a manual check, and a small number of wrong labels surely remain. The rewordings were written by a language model and filtered automatically.

6 Conclusion

We followed the state from which four language models answer short factual questions, layer by layer. In all four, the answer becomes readable from the state within a single layer, at a depth that differs between models. From that layer on, questions with the same answer gather around a shared centroid. The centroid follows the answer the model gives: it does not depend on the type of fact or the wording, it can be read without the directions that write the answer word, and it does not follow the true answer when the model is wrong. The model uses it: moving a state to an answer's centroid after the onset makes the model give that answer, and the same move before the onset does nothing. Because wrong answers have centroids too, the state shows which mistakes are near misses, better than the model's confidence does. Whether the same picture holds for open-ended answers, for reasoning over several steps and for larger models is the natural next question.

Code and data. Code, the dataset and all derived results: https://github.com/gut-puncture/The-Shape-of-Wisdom.

Acknowledgements. This work was inspired by Goodfire's Neural Geometry series [11]. The questions come from Wikidata [36]. Language models (Claude, GPT and Fable) helped with literature review, code review and drafting; all experiments, numbers and claims were checked against the recorded data.

References

[1] Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting Latent Predictions from Transformers with the Tuned Lens. *arXiv preprint arXiv:2303.08112*, 2023.

[2] Chi Seng Cheang, Hou Pong Chan, Wenxuan Zhang, and Yang Deng. Do LLMs Really Know What They Don't Know? Internal States Mainly Reflect Knowledge Recall Rather Than Truthfulness. In Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens, editors, *Findings of the Association for Computational Linguistics: ACL 2026*, pages 713–730, San Diego, California, United States, July 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-acl.34. URL https://aclanthology.org/2026.findings-acl.34/.

[3] Emily Cheng, Diego Doimo, Corentin Kervadec, Iuri Macocco, Jade Yu, Alessandro Laio, and Marco Baroni. Emergence of a High-Dimensional Abstraction Phase in Language Transformers. *arXiv preprint arXiv:2405.15471*, 2024. ICLR 2025.

[4] Róbert Csordás, Christopher D. Manning, and Christopher Potts. Do Language Models Use Their Depth Efficiently? *arXiv preprint arXiv:2505.13898*, 2025. NeurIPS 2025.

[5] Joshua Engels, Eric J. Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. Not All Language Model Features Are One-Dimensionally Linear. *arXiv preprint arXiv:2405.14860*, 2024. ICLR 2025.

[6] Zorik Gekhman, Eyal Ben David, Hadas Orgad, Eran Ofek, Yonatan Belinkov, Idan Szpektor, Jonathan Herzig, and Roi Reichart. Inside-Out: Hidden Factual Knowledge in LLMs. *arXiv preprint arXiv:2503.15299*, 2025. COLM 2025.

[7] Gemma Team et al. Gemma 2: Improving Open Language Models at a Practical Size. *arXiv preprint arXiv:2408.00118*, 2024.

[8] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer Feed-Forward Layers Are Key-Value Memories. *arXiv preprint arXiv:2012.14913*, 2020. EMNLP 2021.

[9] Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. Dissecting Recall of Factual Associations in Auto-Regressive Language Models. *arXiv preprint arXiv:2304.14767*, 2023. EMNLP 2023.

[10] Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A Unifying Framework for Inspecting Hidden Representations of Language Models. *arXiv preprint arXiv:2401.06102*, 2024. ICML 2024.

[11] Goodfire. The Neural Geometry Series. Goodfire Research, 2026. URL https://www.goodfire.com/research/neural-geometry.

[12] Aaron Grattafiori et al. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*, 2024.

[13] Akshat Gupta, Jay Yeung, Gopala Anumanchipalli, and Anna Ivanova. How Do LLMs Use Their Depth? *arXiv preprint arXiv:2510.18871*, 2025.

[14] Wes Gurnee and Max Tegmark. Language Models Represent Space and Time. *arXiv preprint arXiv:2310.02207*, 2023. ICLR 2024.

[15] Wes Gurnee, Emmanuel Ameisen, Isaac Kauvar, Julius Tarng, Adam Pearce, Chris Olah, and Joshua Batson. When Models Manipulate Manifolds: The Geometry of a Counting Task. *arXiv preprint arXiv:2601.04480*, 2026.

[16] Danny Halawi, Jean-Stanislas Denain, and Jacob Steinhardt. Overthinking the Truth: Understanding how Language Models Process False Demonstrations. *arXiv preprint arXiv:2307.09476*, 2023. ICLR 2024.

[17] Hangfeng He and Weijie J. Su. A Law of Data Separation in Deep Learning. *arXiv preprint arXiv:2210.17020*, 2022. PNAS 120(36), 2023.

[18] Jett Janiak, Jacek Karwowski, Chatrik Singh Mangat, Giorgi Giglemiani, Nora Petrova, and Stefan Heimersheim. Characterizing stable regions in the residual stream of LLMs. *arXiv preprint arXiv:2409.17113*, 2024. SciForDL workshop, NeurIPS 2024.

[19] Sunwoo Kim, Haneul Yoo, and Alice Oh. On the Effect of Uncertainty on Layer-wise Inference Dynamics. *arXiv preprint arXiv:2507.06722*, 2025. Actionable Interpretability workshop, ICML 2025.

[20] Vedang Lad, Jin Hwa Lee, Wes Gurnee, and Max Tegmark. The Remarkable Robustness of LLMs: Stages of Inference? *arXiv preprint arXiv:2406.19384*, 2024.

[21] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-Time Intervention: Eliciting Truthful Answers from a Language Model. *arXiv preprint arXiv:2306.03341*, 2023. NeurIPS 2023.

[22] Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories. *arXiv preprint arXiv:2212.10511*, 2022. ACL 2023.

[23] Samuel Marks and Max Tegmark. The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets. *arXiv preprint arXiv:2310.06824*, 2023. COLM 2024.

[24] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and Editing Factual Associations in GPT. *arXiv preprint arXiv:2202.05262*, 2022. NeurIPS 2022.

[25] Giorgos Nikolaou, Tommaso Mencattini, Donato Crisostomi, Andrea Santilli, Yannis Panagakis, and Emanuele Rodolà. Language Models are Injective and Hence Invertible. *arXiv preprint arXiv:2510.15511*, 2025.

[26] nostalgebraist. interpreting GPT: the logit lens. LessWrong, August 2020. URL https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

[27] Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs Know More Than They Show: On the Intrinsic Representation of LLM Hallucinations. *arXiv preprint arXiv:2410.02707*, 2024. ICLR 2025.

[28] Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C. Wallace, and David Bau. Future Lens: Anticipating Subsequent Tokens from a Single Hidden State. *arXiv preprint arXiv:2311.04897*, 2023. doi: 10.18653/v1/2023.conll-1.37. CoNLL 2023.

[29] Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via Contrastive Activation Addition. *arXiv preprint arXiv:2312.06681*, 2023. ACL 2024.

[30] Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The Geometry of Categorical and Hierarchical Concepts in Large Language Models. *arXiv preprint arXiv:2406.01506*, 2024. ICLR 2025.

[31] Yoonah Park, Haesung Pyun, and Yohan Jo. Bridging the Knowledge-Prediction Gap in LLMs on Multiple-Choice Questions. *arXiv preprint arXiv:2509.23782*, 2025. ICML 2026.

[32] Qwen et al. Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115*, 2024.

[33] Valeria Ruscio and Keiran Thompson. On the Tip of the Tongue: Why LLMs Hallucinate Answers They Can Decode. *arXiv preprint arXiv:2603.13911*, 2026.

[34] Mrinank Sharma et al. Towards Understanding Sycophancy in Language Models. *arXiv preprint arXiv:2310.13548*, 2023. ICLR 2024.

[35] Oscar Skean, Md Rifat Arefin, Dan Zhao, Niket Patel, Jalal Naghiyev, Yann LeCun, and Ravid Shwartz-Ziv. Layer by Layer: Uncovering Hidden Representations in Language Models. *arXiv preprint arXiv:2502.02013*, 2025. ICML 2025.

[36] Denny Vrandečić and Markus Krötzsch. Wikidata: a free collaborative knowledgebase. *Communications of the ACM*, 57 (10):78–85, 2014. doi: 10.1145/2629489.

[37] Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. Do Llamas Work in English? On the Latent Language of Multilingual Transformers. *arXiv preprint arXiv:2402.10588*, 2024. ACL 2024.

[38] Sarah Wiegreffe, Oyvind Tafjord, Yonatan Belinkov, Hannaneh Hajishirzi, and Ashish Sabharwal. Answer, Assemble, Ace: Understanding How LMs Answer Multiple Choice Questions. *arXiv preprint arXiv:2407.15018*, 2024. ICLR 2025.

[39] Hugh Mee Wong, Rick Nouwen, and Albert Gatt. When Models Decide and When They Bind: A Two-Stage Computation for Multiple-Choice Question-Answering. *arXiv preprint arXiv:2601.03914*, 2026.

[40] Robert Wu and Vardan Papyan. Linguistic Collapse: Neural Collapse in (Large) Language Models. *arXiv preprint arXiv:2405.17767*, 2024. NeurIPS 2024.

[41] Zhaofeng Wu, Xinyan Velocity Yu, Dani Yogatama, Jiasen Lu, and Yoon Kim. The Semantic Hub Hypothesis: Language Models Share Semantic Representations Across Languages and Modalities. *arXiv preprint arXiv:2411.04986*, 2024. ICLR 2025.

[42] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large Language Models Are Not Robust Multiple Choice Selectors. *arXiv preprint arXiv:2309.03882*, 2023. ICLR 2024.

A Details

Questions. Subjects were sampled from three popularity bands by number of Wikipedia language editions (1–4, 5–19, and 20 or more), roughly a third from each, because models know popular facts far better than rare ones [22]. Every subject has one answer, a unique English name and no namesake of the same kind. We dropped Hindi as the language a person spoke after a manual check found its labels unreliable. A rewording was kept only if it named the subject exactly, asked a question, mentioned no candidate answer or its demonym, and differed from the plain question; 4,323 passed. Multiple-choice questions show four options in all four rotations of their order. In total each model answers 15,603 prompts, each in its own chat format.

States and grading. States are recorded before the final normalisation, exactly as the model computes them in bfloat16, at the last prompt position before the first layer and after every layer. A reply is graded by its first line, matched against hand-curated names for each answer (for example “China” or “Mandarin”). A single answer outside the 62 candidates counts as wrong.

Centroids and readability. A centroid is built only for an answer given to at least five training questions about at least three different subjects. Readability counts only test questions whose given answer has a centroid. The whitening keeps the 128 leading principal components of the training states at each layer and divides each by its standard deviation, after shrinking each variance 10% toward the average variance so that directions with very little variance are not blown up. When a result is reported at a given depth, we use the nearest layer.

Onset layer. We split the training subjects in half again, build centroids from one half, and measure readability on the other at every layer. The onset layer is the first layer at which this readability covers half the distance from chance to its highest value over layers, where chance is the average, over questions, of one over the number of candidate answers of the question’s kind.

Uncertainty. Questions about one subject are not independent, so we resample subjects: we draw the test subjects with replacement 300 times, recompute the result each time, and report the middle 95% of the results.

Controls. Shuffled answers are permuted between facts of the same kind, all wordings of a fact keeping one answer; readability is averaged over five shuffles. For the question-text control, the TF-IDF vectors of character sequences of length two to four are reduced to 256 dimensions by a truncated singular value decomposition fitted on training questions.

Separation. D_{between} and D_{within} are computed for each kind of answer separately and each is summed over kinds, weighting each kind by its number of test questions, before taking their ratio.

Centroids follow the answer. For other relations, each test question’s readability is averaged over the centroids built from each other relation of its kind. For answer words removed, each unembedding vector is first multiplied by the final normalisation’s scale, the vectors are turned into an orthonormal basis, and each state is projected onto the space at right angles to that basis; the whitening is then refitted. The random control removes a random orthonormal basis of the same size. True-answer centroids are built like ordinary centroids but group training questions by their true answer. The cross-model comparison uses plain test questions at each model’s onset layer.

Moves. The seven depths are 20%, 35%, 50%, 60%, 70%, 80% and 90%, each rounded to the nearest layer. The move is added once, at the last prompt position, while the model reads the prompt; generation then proceeds without changes. The third answer, and the other answer for right answers, are drawn at random from answers of the same kind that have a centroid. Right answers are a random sample of 200 per model. Wrong answers whose given answer has no centroid are excluded from the numbers in Section 3.3.

Near misses. The rank of the right centroid is computed in whitened states, for wrong answers whose right answer has a centroid; both numbers are computed on these same wrong answers. As a check, logistic regressions on three geometric features (distance to the right centroid, its rank, and the margin between the two nearest centroids) and on the reply’s log-probability and length, scored by five-fold cross-validation with each subject kept in one fold, give AUC 0.73–0.83 against 0.50–0.69.